

The 5 Projects That Actually Get You Hired As A Quant in 2026

Every candidate knows Black-Scholes and basic Bayes theorem.

Top buyside desks filter 95% of resumes. They hire production builders.

Here are the 5 architectures that prove you are desk-ready on Day 1:

01 // HFT & C++20

L2 Limit Order Book & Matching Engine

- Lock-free SPSC circular queues
- Probabilistic queue depletion

02 // VOLATILITY

Arbitrage-Free SVI Dupire Local Vol Surface

- Gatheral SSVI parameterization
- Breeden-Litzenberger density

03 // STAT ARB

Kalman Filter Pairs Dynamic Beta Stat Arb

- Cointegration: ADF & Johansen
- Ornstein-Uhlenbeck half-life

04 & 05 // RATES & RISK

SOFR Multi-Curve & XVA American MC

- Dual-curve OIS bootstrapping
- Longstaff-Schwartz EE / PFE

SWIPE FOR ARCHITECTURES & INTERVIEWS →

L2 Limit Order Book & Matching

Language: Modern C++20 | Focus: Microstructure & Ultra-Low Latency

AMATEUR APPROACH (AUTO-REJECTED)

"Downloaded daily AAPL bars with yfinance, backtested 20/50 SMA crossover."

Why it gets rejected: Citadel and Optiver auto-reject this in 5 seconds. Shows zero proof of order matching, queue priority, exchange latency, or adverse selection.

PRODUCTION DESK STANDARD (BUY-SIDE)

What top market making desks look for in your codebase:

- Price-Time Priority Engine: Doubly linked lists with $O(1)$ order cancels via hash map
- Lock-Free SPSC Ring Buffer: Cache-line aligned circular queues (zero mutex contention)
- Probabilistic Queue Position: Model volume ahead of you; never assume naive touch fills
- Microsecond Latency Injector: Pessimistic wire-to-wire ack offsets to reflect live PnL
- Synthetic ITCH / OUCH Parser: Replay high-burst market events with zero packet drops

CV Impact: Proves you can build low-latency systems that survive production traffic.

DESK INTERVIEW DEFENSE QUESTION

"Your limit order sits at the inside bid. 40 lots trade at the bid. Did your order get filled? How does your engine track queue priority?"

Your project codebase gives you verified 10-second proof during technical rounds.

Arbitrage-Free SVI Vol Surface

Language: Python + Modern C++ | Focus: Dupire Local Vol & Static Arbitrage

AMATEUR APPROACH (AUTO-REJECTED)

"Scraped Yahoo Finance option chains and fitted cubic splines across implied vol."

Why it fails: Standard splines violate no-arbitrage bounds, producing negative butterfly probability densities and imaginary Dupire local volatilities.

PRODUCTION DESK STANDARD (BUY-SIDE)

What volatility desks (Citadel, Goldman, Millennium) demand:

- Gatheral Quasi-Explicit SVI / SSVI: Consistent parameterization across strike & maturity
- Calendar Arbitrage Elimination: Enforce strict $dW/dT \geq 0$ across total variance curves
- Breeden-Litzenberger Density: Enforce $d^2C/dK^2 \geq 0$ for positive risk-neutral density
- Dupire Local Vol Inversion: Smooth PDE pricing grid for exotic autocallables & barriers
- Roger Lee Extreme Wing Bounds: Enforce asymptotic slope limits to stop tail explosion

CV Impact: Shows complete mastery of continuous-time stochastic volatility modeling.

DESK INTERVIEW DEFENSE QUESTION

"Under what conditions does SSVI surface calibration produce butterfly arbitrage? How does your optimizer enforce wing limits?"

Direct mathematical derivations and runnable C++ code included in the pack.

Kalman Filter Stat Arb Engine

Language: Python | Focus: Cointegration, OU Mean-Reversion & Frictions

AMATEUR APPROACH (AUTO-REJECTED)

"Calculated Pearson correlation between KO and PEP, traded when Z-score > 2."

Why it fails: Correlation $\neq$ Cointegration. High correlation between non-stationary prices is often spurious. Unhedged spreads drift apart and blow up the portfolio.

PRODUCTION DESK STANDARD (BUY-SIDE)

What quantitative equity & statistical arbitrage teams test for:

- Cointegration Rigor: Augmented Dickey-Fuller (ADF) & Johansen eigenvector trace rank
- State-Space Kalman Filter: Dynamically update beta in real time with transition matrix
- Ornstein-Uhlenbeck (OU) Fitting: Exact analytical half-life calculation for trade sizing
- Almgren-Chriss Market Impact: Square-root slippage model to eliminate phantom alpha
- Structural Break Guards: CUSUM filter halting orders when cointegration breaks down

CV Impact: Proves you design robust econometric alphas that survive real market friction.

DESK INTERVIEW DEFENSE QUESTION

"Why does rolling OLS fail for hedge ratio estimation compared to a Kalman filter? How do you tune process noise covariance Q?"

Step-by-step state-space formulation and math derivations in the pack.

SOFR Multi-Curve & XVA Engine

Modern C++ | Dual-Curve Bootstrapping & American Monte Carlo

PROJECT 04: MULTI-CURVE SOFR

The Post-LIBOR Dual-Curve Architecture:

- Dual-Curve Separation: SOFR/OIS discounting vs. Term SOFR projection curves
- Monotone Convex Splines: Hagan-West interpolation preventing negative forward rates
- Jacobian Solving: Simultaneous multi-instrument root finding for IRS and basis swaps

Desk takeaway: Single-curve discounting is obsolete. Desks test dual-curve mastery.

PROJECT 05: XVA SIMULATION ENGINE

Counterparty Credit Risk Under Longstaff-Schwartz:

- American Monte Carlo (AMC): Longstaff-Schwartz boundary across 10,000 paths
- Exposure Profiles: Expected Exposure (EE) and 99% Potential Future Exposure (PFE)
- CSA Master Agreements: Netting sets, collateral thresholds, and MPOR modeling

Desk takeaway: Risk quants must model netting and default intensity jump curves.

DESK INTERVIEW DEFENSE QUESTION

"How do you calculate CVA on a netting set with daily margin calls vs. an uncollateralized trade? How does your code model WWR?"

Production Longstaff-Schwartz C++ implementation included in the pack.

Don't Just Learn Theory. Build Real Quant Engines.

We compiled all 45 production-grade projects into a complete blueprint:

45 INDUSTRY GRADE PROJECTS

OFFICIAL DESK RELEASE

Ultimate Industry Grade Quant Project Pack

Mathematical Derivations · Python Prototypes · Production C++20

- 45 Full Desk Projects: Pricing, Risk, XVA, Rates, FX & Equity Stat Arb
- Dual-Implementation: Math Derivations to Python Prototypes to C++
- Interview Defense Guide: 90+ technical desk questions with verified answers
- Resume-Ready Bullets: Pre-written impact bullet points for your quant CV
- 600+ Pages of Code: Runnable Jupyter notebooks & production C++ engines

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